August 4, 2026

Dear Hiring Team,

I am writing to apply for the Research Engineer II position, requisition R-0000150239, with RBC Borealis. With a Master of Science in Computer Science from the University of Calgary and applied experience in Python, PyTorch, scalable research software, model evaluation, and data processing, I am excited by the opportunity to contribute to machine learning solutions that move from exploration and prototyping toward real product impact.

My graduate research focused on building machine learning and data-management systems for large geospatial and Earth observation datasets. In my digital elevation model super-resolution project, I designed Python and PyTorch workflows for data collection, preprocessing, model training, and evaluation. I trained deep ResNet models for 4x terrain reconstruction and compared model quality using RMSE, slope, TRI, TPI, and wavelet-based metrics. This work required careful baseline thinking, reproducible experimentation, and honest assessment of whether models generalized beyond convenient random splits.

I also developed C++ and Python software for scalable processing, storage, and retrieval of satellite imagery through my research work with BigGeo, Mitacs, and the University of Calgary. I built DGGS-based encoding pipelines, designed tensor-based storage for multiresolution retrieval, and improved processing performance with C++ multithreading. These projects strengthened the engineering habits I would bring to RBC Borealis: modular implementation, performance awareness, documentation of design decisions, and a focus on making research outputs usable by others.

Beyond my thesis work, I have been building an agentic job-search automation platform using OpenClaw, n8n, LLM workflows, Docker, SearXNG, Google Sheets, Google Drive, and Telegram integrations. The project is helping me work more directly with AI workflow orchestration, search-driven decision support, and tool integration. While my strongest production experience is in backend and data systems, this project reflects my interest in practical machine learning systems that connect models, software, and user-facing workflows.

RBC Borealis appeals to me because the role sits at the intersection of applied research, engineering quality, and collaboration with business and product stakeholders. I would be glad to contribute my Python, PyTorch, data-processing, prototyping, and technical communication background while continuing to grow in areas such as information retrieval, NLP, GenAI, and production machine learning systems.

Thank you for considering my application. I would welcome the opportunity to discuss how my research engineering experience can contribute to RBC Borealis.

Sincerely,
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